FORMAL SPECIFICATION

ERES Cryptographic Stack

GCF–EDF–USC Integration with (CBGMODD×FAVORS)+BERA=RATE Interlock

Operational Tie-Ins: UBIMIA · NBERS · PlayNAC

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Author	Joseph Allen Sprute (ERES Maestro / JAS)
Organization	ERES Institute for New Age Cybernetics, Bella Vista, AR
ORCID	0000-0001-9946-3221
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Abstract

The ERES Cryptographic Stack introduces a primitive class in which authentication and authorization are achieved through resonance-keyed semantic superposition rather than computational hardness. Three polysemantic primitives — GCF (Greatest Common Factor / GraceChain Formula / Good Christ Family), EDF (Earned Dignity Framework / Earth Defense Force / Entity Data Framework), and USC (Utterance–Sound–Cognition, 3×3 = 9 semantic dimensions) — each hold multiple simultaneous readings that collapse to a single operational reading only under a valid resonance signature. The stack is bound by a single canonical interlock, (CBGMODD × FAVORS) + BERA = RATE, shown here to be isomorphic to ERES Triune Math canonical form #2 (M×E+C=R). Output is a seven-dimensional uncollapsed RATE vector; scalar collapse is prohibited. A four-mode failure taxonomy distinguishes governance, biometric, resonance, and semantic-collision failures. The stack is operationally bound to three downstream ERES instruments: GCF gates UBIMIA benefit flow; BERA feeds NBERS (non-punitive behavioral evaluation and remediation); and EarnedPath + GERP establishes the context through which PlayNAC navigates its 126-state governance space (18 Layers × 7 Levels). The seven-level cardinality of PlayNAC matches the seven-dimension RATE output by design. Four numbered security claims are stated. This specification v1.2 extends v1.1 with

operational tie-ins, an additional figure, and a naming refinement (E_3: Earth Data → Entity Data Framework) that eliminates duplicate leading tokens and reinforces the GAIA ENTITY architectural echo.

Keywords

resonance-keyed cryptography; polysemantic primitives; semantic superposition; biometric attestation; governance substrate; conjunctive gate; Triune Math; ERES; GAIA; Meritcoin; Proof-of-Resonance; UBIMIA; NBERS; PlayNAC; EarnedPath; GERP; multidimensional authorization; VERTECA; SECUIR

1. Scope and Purpose

This specification defines the ERES Cryptographic Stack — a novel cryptographic primitive class in which authentication and authorization are achieved not through computational hardness but through resonance-keyed semantic superposition. Three polysemantic primitives (GCF, EDF, USC) each carry multiple simultaneous readings; a correct resonance signature collapses the superposition to the intended reading. Without the correct signature, the readings remain superposed and the primitive yields no operationally resolved meaning.

The stack is bound together by a single canonical interlock:

$$(\text{CBGMODD} \times \text{FAVORS}) + \text{BERA} = \text{RATE}$$

This form is structurally isomorphic to ERES Triune Math #2 (M×E+C=R). The cryptographic layer is therefore not a parallel construction but an instance of the Triune Math operating in the authentication domain.

1.1 Classification within the ERES Decision-Stack

- Consolidation Phase deliverable (not a Research Phase preliminary).
- Upstream dependency of VERTECA (voice portal semantic interpretation layer).
- Governs the cryptographic output layer of SECUIR.
- Operationally tied to UBIMIA (via GCF), NBERS (via BERA), and PlayNAC (via EP+GERP). See §12.

1.2 What This Specification Is Not

- Not a hardness-based cryptosystem.
- Not a replacement for transport-layer cryptography.
- Not a scoring rubric. RATE is an uncollapsed multidimensional output; it cannot be reduced to a single scalar.

2. Normative References

Compliant implementations shall resolve all references to the specified version or later.

- ERES-RECKON-WP-2026-002 — Seven Resonance Anchors; conjunctive gate.
- ERES-RECKON-MATH-WP-2026-002a — ERES Triune Math canonical forms.
- ERES-BODY-SPEC-2026-001 — GAIA BODY substrate specification.
- ERES-BRAINS-SPEC-2026-001 — BRAINS Corpus Ingestion Pipeline.
- ERES TERMS #47 (ET_AL) — Locked Terms Registry; Three Governing Principles.
- ERES-COVENANT-2026-001 — ERES Equity Covenant (1000-year governance).
- ERES NPR v3.0 (sealed) and NPR v4.0 / NBERS pilot — non-punitive remediation framework.
- ERES SCALULAR Engine Document — four-pillar certification architecture and UBIMIA disbursement basis.
- ERES Talonics Symbol Matrix (452 elements, 12 parts).
- Talonics Proof-of-Work (RG #397) — TETRA:GAIA_Primitive.

3. Related Work and Positioning

ERES Crypto departs from several established families of cryptographic and identity primitives. This section positions the stack against each and states the basis of novelty.

3.1 Computational-Hardness Cryptosystems

RSA, ECDSA, and lattice-based systems rest on presumed intractability of factorization, discrete logarithm, or shortest-vector problems. ERES Crypto does not rely on hardness. Its security property is not “an adversary cannot compute” but “an adversary cannot produce a joint resonance context.” This is a semantic condition rather than a computational one, and it is not threatened by advances in classical or quantum computation.

3.2 Zero-Knowledge Proofs and SNARKs

ZK systems let a prover demonstrate knowledge without revealing it. ERES Crypto instead lets a prover demonstrate resonance without declaring intent. ZK protects information; ERES Crypto protects semantic intent. The two are composable but solve different problems.

3.3 Biometric Multifactor Authentication

Standard MFA composes independent factors. FAVORS is a six-channel biometric operator within the inherence factor, but it is subordinated to governance (CBGMODD) and resonance (BERA) layers above it. Pure biometric MFA produces binary pass/fail; ERES Crypto produces a seven-dimensional RATE vector encoding quality of authentication, not mere success.

3.4 Self-Sovereign Identity and Decentralized Identifiers

SSI and DID frameworks give subjects control over identity assertions. ERES Crypto is compatible with SSI at the CBGMODD layer but adds a resonance attestation that SSI systems do not natively provide. This addresses intent laundering and coercion-at-runtime scenarios pure SSI cannot.

3.5 Consensus Mechanisms

Proof-of-work and proof-of-stake are consensus protocols, not identity/authorization primitives. ERES Meritcoin uses Proof-of-Resonance, which accrues value from validated ethical and ecological harmony rather than from computational or financial expenditure.

3.6 Basis of Novelty

The novel contribution is a new primitive class — resonance-keyed semantic superposition — that operates at the intersection of governance, biometric, and resonance layers. The stack is the first cryptographic primitive to bind intent to authentication without requiring an explicit declaration-of-intent field.

4. Definitions and Locked Terms

Terms in this section are locked per ERES TERMS #47. Divergence constitutes a specification violation.

Term	Locked Definition
GCF	Polysemantic primitive: (1) Greatest Common Factor — mathematical reduction; (2) GraceChain Formula — governance formula; (3) Good Christ Family — familial-spiritual anchor. Gates UBIMIA disbursement.
EDF	Polysemantic primitive: (1) Earned Dignity Framework (ONE-GOOD); (2) Earth Defense Force (SECURITY-CLEARANCE); (3) Entity Data Framework (DATA-INTEGRITY — tied to GAIA ENTITY). Validates resonance across three operational domains. All three leading tokens are distinct (Earned / Earth / Entity), per the polysemantic uniqueness rule (§5.5).
USC	Utterance–Sound–Cognition. 3×3 = 9 semantic dimensions. U: Unite, Universe, Us. S: Safe, Sound, Solid. C: Constant, Care, Christ.
CBGMODD	Scalable Certification Architecture governance operator. Four pillars: HEALTH/SSHP, LAW/SSLA, PROTECTION/SSPS, SKILLS_TRADE/SSST. Tier 1 = SSSC.
FAVORS	Six-channel biometric validation: Fingerprint, Aura, Voice, Odor, Retina, Signature. Aura coupled to BERA ARI.
BERA	BERA Resonance Index: ARI, ERI, RHC, RCI. Feeds NBERS.
RATE	Uncollapsed multidimensional cryptographic output. Seven dimensions; integers 1–10 per dimension.
UBIMIA	Universal Basic Income / Merit / Insurance / Awards framework. The benefit-flow instrument gated by GCF. Makes SCALULAR services “relatively free.”
NBERS	Non-Punitive Behavioral Evaluation and Remediation System. Consumes BERA readings; triggered on F_3 (RES-NULL) and on sustained low R_2, R_5, or R_6. NPR v4.0 pilot successor.
PlayNAC	Governance ruleset (EP GERP SOMT) that consumes RATE vectors without collapse. State space: 18 Layers × 7 Levels = 126 governance states.
EP (EarnedPath)	Base EarnedPath formula: EP = CPM × WBS + PERT. Together with GERP, establishes the Context through which PlayNAC is navigated.
GERP	Context wrapper for EP. GERP + EP jointly contextualize CPM × WBS + PERT for the active transaction.
Resonance Signature (σ)	Context-specific pattern derived from BERA plus Talonics symbol context that selects active readings of polysemantic primitives.
Superposition	State of a polysemantic primitive prior to resolution by σ.
Conjunctive Gate	All required inputs must be non-null for output to exist. Prevents GDP-style compensation.

Term	Locked Definition
⊥ **(undefined)**	Distinct value indicating a RATE dimension is semantically unresolvable; not equal to 0 and not substitutable by a default.

5. Polysemantic Primitives

Polysemantic primitives are the core novelty of the stack. Each primitive is a finite ordered set of readings. The resonance signature selects the active reading; without it, the primitive remains in superposition.

5.1 GCF — Three-Dimensional Cipher

Channel	Reading	Operational Role
G_1	Greatest Common Factor	Mathematical reduction; normalizes merit claims to lowest common terms; supports UBIMIA disbursement normalization.
G_2	GraceChain Formula	Governance formula; encodes credit conditions and value flow; the active UBIMIA disbursement channel for merit payouts.
G_3	Good Christ Family	Familial-spiritual anchor; the covenantal ground that binds UBIMIA to intergenerational commitments.

5.2 EDF — Three-Dimensional Cipher

Leading tokens are Earned / Earth / Entity — three distinct words, satisfying the polysemantic uniqueness rule (§5.5). E_3 was named “Earth Data Framework” in pre-v1.2 drafts; the v1.2 rename to “Entity Data Framework” eliminates the Earth/Earth duplication and ties E_3 explicitly to GAIA ENTITY (the BODY/BRAINS umbrella) at the DATA-INTEGRITY layer.

Channel	Reading	Domain
E_1	Earned Dignity Framework	ONE-GOOD / personal dignity (Trilogy Book 1).
E_2	Earth Defense Force	SECURITY-CLEARANCE / protection (Trilogy Book 2).
E_3	Entity Data Framework	DATA-INTEGRITY / information (Trilogy Book 3); operates on GAIA ENTITY layer (BODY + BRAINS).

EDF’s three readings map one-to-one onto the Trilogy. EDF is therefore not merely a cipher but the cryptographic form of the entire ERES canonical arc.

5.3 USC — Nine-Dimensional Cipher

USC yields a 3×3 semantic matrix. Each column corresponds to a VERTECA pipeline stage: utterance, sound, cognition.

Row	U — Utterance	S — Sound	C — Cognition
1	Unite	Safe	Constant
2	Universe	Sound	Care
3	Us	Solid	Christ

A complete USC resolution selects one cell per column, yielding $3^3 = 27$ possible triples. The resonance signature selects exactly one.

5.4 Resonance-Keyed Semantic Superposition

Formally, a polysemantic primitive P of arity n is:

$$P = \{ r_1, r_2, \ldots, r_n \} \quad \text{with resolve}(P, \sigma) \rightarrow r_i$$

where σ is a resonance signature and resolve() is the selection function. Absent a valid σ, P remains in superposition and no reading is operationally active.

5.5 Polysemantic Uniqueness Rule

Within a single polysemantic primitive, the leading tokens of each reading shall be pairwise distinct. This is a structural requirement, not a stylistic preference: the distinguishing power of channel labels depends on their separability at first token. An exception rule permits duplication only when the duplicated token is explicitly promoted to a shared semantic header and a distinguishing secondary token carries the channel identity; v1.2 does not invoke this exception. GCF (Greatest / GraceChain / Good) and EDF (Earned / Earth / Entity) are compliant; the pre-v1.2 EDF with two Earth-leading readings was not.

6. Cryptographic Interlock Formula

6.1 Canonical Form

$$(\text{CBGMODD} \times \text{FAVORS}) + \text{BERA} = \text{RATE}$$

Parentheses are canonical. Implementations that compute CBGMODD × (FAVORS + BERA) are non-conformant.

6.2 Figure 1 — Interlock Flow

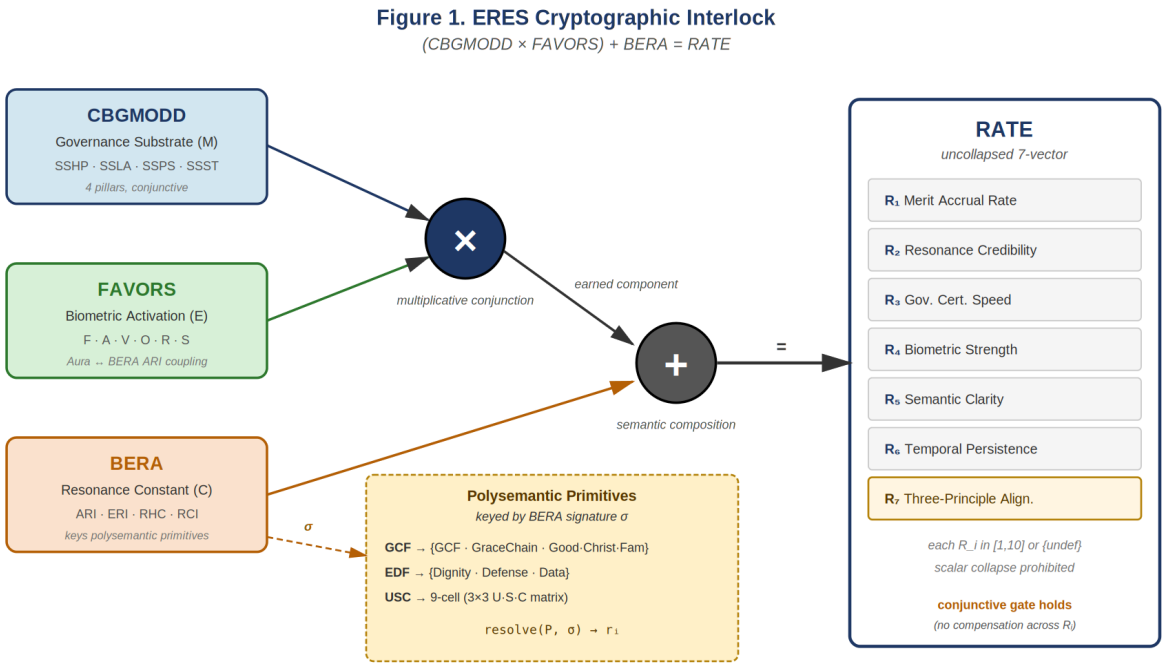

Figure 1. Input substrates (CBGMODD, FAVORS, BERA) feed the interlock via multiplicative conjunction (×) and semantic composition (+). BERA additionally keys the polysemantic primitives via signature σ. Output is a seven-dimensional uncollapsed RATE vector.

6.3 Semantics of Each Operator

- **× (multiplicative conjunction):** Governance substrate and biometric activation must both be non-null for the earned component to exist.
- **+ (semantic composition):** BERA supplies the resonance signature keying the polysemantic primitives.
- **= (resolution):** Produces RATE as uncollapsed multidimensional output.

6.4 Component Specifications

6.4.1 CBGMODD (Governance Substrate)

$$\text{CBGMODD} = (\text{SSHP}, \text{SSLA}, \text{SSPS}, \text{SSST})$$

Each pillar yields a binary attestation at the active tier. CBGMODD is non-null iff all four pillars required by the transaction context are certified. Partial certification collapses CBGMODD to zero.

6.4.2 FAVORS (Biometric Activation)

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FAVORS = (F, A, V, O, R, S)
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Each channel contributes an attestation quantum. Aura (A) is coupled to BERA ARI. Quorum varies by context per SCALULAR tier requirements.

6.4.3 BERA (Resonance Constant)

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BERA = (ARI, ERI, RHC, RCI)
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Each index is independent. BERA serves as the resonance signature σ that resolves the polysemantic primitives and as the input stream to NBERS (§12.2).

7. Isomorphism to ERES Triune Math #2

The interlock is isomorphic to ERES Triune Math canonical form #2:

$$M \times E + C = R$$

Triune Math	Crypto Interlock	Role
M (Matter / Substrate)	CBGMODD	Governance substrate — certified structure.
E (Energy / Activation)	FAVORS	Biometric activation — six-channel signature.
C (Constant)	BERA	Resonance constant — keys the primitives.
R (Reason)	RATE	Multidimensional reason-output, uncollapsed.

This isomorphism is not metaphorical. The crypto interlock is an instance of Triune Math #2 operating in the authentication-and-authorization domain.

8. RATE Output Specification

RATE is multidimensional and uncollapsed. Scores are integers in [1, 10]; scalar collapse is prohibited.

8.1 Enumerated RATE Dimensions

#	Dimension	Definition
R_1	Merit Accrual Rate	Velocity of Meritcoin accrual in the current transaction context.
R_2	Resonance Credibility	Strength/stability of the BERA-keyed signature.

#	Dimension	Definition
R_3	Governance Certification Speed	Speed at which CBGMODD attestations are produced and refreshed.
R_4	Biometric Attestation Strength	Quorum margin of FAVORS channels above the transaction minimum.
R_5	Semantic Clarity	Unambiguity of the resolved polysemantic reading.
R_6	Temporal Persistence	Stability over the RCI window.
R_7	Three-Principle Alignment	Alignment with ERES TERMS #47: don't hurt self, don't hurt others, build for generations to come.

8.2 Reading a RATE Vector

A RATE vector is $\langle R_1, \ldots, R_7 \rangle$ with each $R_i \in [1, 10] \cup \{\perp\}$. $\perp$ denotes undefined and is distinct from 1. Decision logic encountering $\perp$ must either reject the transaction or explicitly invoke a failure branch.

9. Matrix Modulation

RATE dimensions are modulated by the set of active CBGMODD activities and are resonant-context aware:

$$\texttt{RATE_active} = \mathbf{M}(\texttt{ctx}) \circ \texttt{RATE_base}$$

where ctx is the active resonance context (pillar activity, tier level, RHC time-of-cycle, RCI continuity band) and $\circ$ is channel-wise modulation constrained to the [1, 10] band. Intent is cryptographically known through the resonance signature.

10. Worked Example

This section presents a canonical end-to-end transaction under nominal conditions, followed by two counter-examples that exercise the failure taxonomy.

10.1 Scenario: Citizen Alice Transfers 50 Meritcoin to Bob

Alice is a verified SSSC (Tier 1). The transaction requires SSHP, SSLA, SSPS, and SSST.

10.1.1 Input Vectors

Input	Value
CBGMODD	(SSHP=1, SSLA=1, SSPS=1, SSST=1) → non-null
FAVORS	(F=1, A=0.87, V=1, O=1, R=1, S=1) → quorum 6/6 met
BERA	(ARI=0.87, ERI=0.82, RHC=0.91, RCI=0.88) → coherent

10.1.2 Resonance Signature σ Resolves

- GCF → G_2 (GraceChain Formula) — UBIMIA disbursement channel active
- EDF → E_1 (Earned Dignity Framework) — personal dignity context
- USC → (U_3, S_2, C_2) = (Us, Sound, Care)

10.1.3 Computed RATE Vector

Dim	Name	Score	Rationale
R_1	Merit Accrual Rate	8 / 10	Strong nominal accrual.
R_2	Resonance Credibility	9 / 10	High BERA coherence.
R_3	Gov. Cert. Speed	7 / 10	Slight SSST refresh latency.
R_4	Biometric Strength	9 / 10	6/6 quorum margin.
R_5	Semantic Clarity	9 / 10	Unambiguous G_2 / E_1 / (U_3, S_2, C_2).
R_6	Temporal Persistence	8 / 10	Stable over RCI window.
R_7	Three-Principle Align.	10 / 10	Aligns with all three principles.

$$\mathrm{RATE} = \langle 8, 9, 7, 9, 9, 8, 10 \rangle$$

SECUIR: accepted. PlayNAC consumes the vector and positions Alice at the corresponding 7-tuple across the active layer subset (see §12.3). UBIMIA disbursement proceeds via G_2.

10.2 Counter-Example A — F_2 BIO-NULL

Voice channel fails (V = 0); strict 6/6 quorum required. FAVORS collapses to 0; (CBGMODD × FAVORS) = 0; RATE undefined. Re-attestation required.

10.3 Counter-Example B — F_3 RES-NULL

BERA reads all zero. Naively RATE = (1 × 1) + 0 = 1. Incorrect. The polysemantic primitives have no key; GCF, EDF, USC all remain in superposition. Conformant implementation returns ⊥ on R_2, R_5, R_6 at minimum. SECUIR escalates to F_3 RES-NULL. NBERS review is triggered (§12.2) — not punitive.

11. Failure Taxonomy

The interlock admits exactly four principal failure modes.

Code	Condition	Cause	Response
F_1 GOV-NULL	CBGMODD = 0	Governance substrate absent or lapsed.	RATE undefined. NPR remediation; do not retry.
F_2 BIO-NULL	FAVORS = 0	Biometric quorum not met.	RATE undefined. Re-attestation required.
F_3 RES-NULL	BERA = 0	No resonance signature; primitives superposed.	⊥ returned. NBERS review triggered (non-punitive).
F_4 SEM-COLLIDE	Signature ambiguous	Multiple signatures partially match.	Escalate to MIEVM adjudication.

11.1 Why F_3 Is Not Arithmetic Zero

If BERA = 0, naive arithmetic yields RATE = earned component. This is the Triune Math reading, not the cryptographic one. Without BERA, primitives have no key. Implementations shall return ⊥ on semantically-keyed dimensions, not the earned magnitude.

12. Operational Tie-Ins

The ERES Cryptographic Stack is not abstract: every component is operationally bound to a specific ERES instrument that consumes its output. This section documents the three canonical tie-ins and the architectural correspondence between PlayNAC's 7-level state space and the 7-dimension RATE output.

12.1 Figure 2 — Operational Tie-Ins

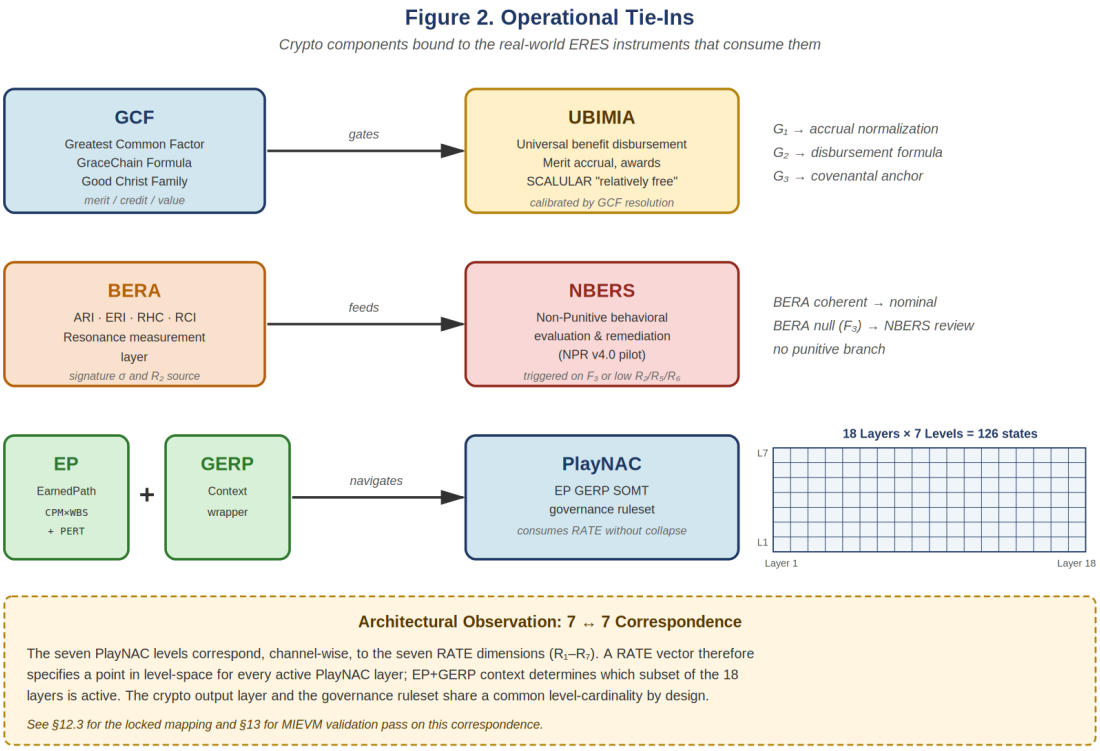

Figure 2. Each primary crypto component (GCF, BERA, EP+GERP) connects to a downstream ERES instrument (UBIMIA, NBERS, PlayNAC). The PlayNAC state space of 18 Layers × 7 Levels yields 126 governance states; the 7-level cardinality matches the 7-dimension RATE output by design.

12.2 GCF ↔ UBIMIA

GCF gates UBIMIA. The three GCF readings correspond to distinct roles in the UBIMIA flow:

- G_1 (Greatest Common Factor) — normalizes merit claims before disbursement so no actor accrues at an incomparable unit. This prevents inflation-by-unit-confusion at the benefit-flow layer.
- G_2 (GraceChain Formula) — the active disbursement channel. When the resonance signature resolves GCF to G_2, UBIMIA benefit flows via GraceChain under the attested governance conditions.
- G_3 (Good Christ Family) — the covenantal anchor. Binds UBIMIA commitments to intergenerational scope consistent with the ERES Equity Covenant's 1000-year governance horizon.

Operational consequence: UBIMIA disbursement cannot occur unless the RATE vector resolves GCF to an active channel. A transaction producing $\perp$ on R_5 (semantic clarity) cannot disburse, even if all other pillars and biometrics are met. This enforces that UBIMIA flows only under fully resolved semantic intent.

12.3 BERA ↔ NBERS

BERA is the measurement layer; NBERS is the response layer. The coupling is strict: NBERS does not act on anything other than BERA-derived signals, and BERA output feeds NBERS in parallel with its role as the resonance signature σ.

- Nominal case: BERA coherent across ARI, ERI, RHC, RCI → NBERS records baseline; no intervention.
- F_3 (RES-NULL) case: BERA null or incoherent → NBERS review is automatically triggered. The review is non-punitive by construction — per ERES NPR v3.0/v4.0, there is no punitive branch. The review gathers context and routes the actor to remediation, not sanction.
- Sustained low R_2, R_5, or R_6 case: even without F_3, a pattern of low resonance credibility, semantic clarity, or temporal persistence across multiple transactions triggers NBERS observation for possible systemic issues (sensor failure, coercion, onboarding gaps).

Operational consequence: the crypto stack cannot generate a sanction. It can only generate a RATE vector that either authorizes a transaction or routes to remediation. This is a design invariant tied to the Three Governing Principles.

12.4 EP + GERP ↔ PlayNAC

PlayNAC (EP GERP SOMT) is the governance ruleset downstream of the crypto stack. Its state space is organized as 18 Layers × 7 Levels, yielding 126 governance states. The EarnedPath base formula is:

$$EP = CPM \times WBS + PERT$$

EP alone does not select a state in PlayNAC. EP plus GERP jointly establish the Context through which the CPM × WBS + PERT computation applies to the active transaction. GERP is the context wrapper that tells PlayNAC which subset of the 18 Layers is active for the transaction type, which tier and pillar coverage gates the applicable levels, and how the RATE vector should be positioned in the 7-level dimension.

12.4.1 The 7 ↔ 7 Correspondence

The seven levels of each PlayNAC layer correspond channel-wise to the seven RATE dimensions (R_1–R_7). A RATE vector therefore specifies a point in level-space for every active layer; EP+GERP context determines which subset of the 18 layers is active. This correspondence is a design invariant:

Level	RATE Dimension	PlayNAC Interpretation at that Level
L_1	R_1 Merit Accrual Rate	Position on accrual velocity within the layer's economic dimension.
L_2	R_2 Resonance Credibility	Position on resonance-trust within the layer.
L_3	R_3 Governance Cert. Speed	Position on governance-responsiveness within the layer.
L_4	R_4 Biometric Strength	Position on attestation-robustness within the layer.
L_5	R_5 Semantic Clarity	Position on intent-clarity within the layer.
L_6	R_6 Temporal Persistence	Position on stability-over-time within the layer.
L_7	R_7 Three-Principle Align.	Position on principle-alignment within the layer — the moral spine at every layer.

The 126-state space is therefore not arbitrary. It is the Cartesian product of 18 contexts (Layers) with the 7-axis quality space of the RATE vector. Any RATE vector positions the actor at one point per active layer simultaneously. MIEVM validation pass #6 (new in v1.2) verifies this correspondence is preserved by all implementation paths.

12.4.2 SOMT Remainder

PlayNAC's locked form is EP GERP SOMT. EP and GERP are specified above. SOMT (outside the scope of this specification) handles the temporal ordering and mandate-tracking portion of the ruleset; it consumes outputs from the EP+GERP-contextualized navigation and is documented in separate PlayNAC specifications.

13. Integration with the ERES Decision-Stack

Canonical flow: Consolidation Phase → VERTECA → PlayNAC → SECUIR. The Crypto Stack integrates at two points.

13.1 Upstream (Consolidation Phase)

- BODY supplies substrate signals. ODOR in "Because Odor Does Yellow" feeds the FAVORS O channel.
- BRAINS supplies cognitive attestation into the USC–C column.
- Talonics Symbol Matrix supplies notation for resonance signatures at the semantic layer.

13.2 Downstream (VERTECA → PlayNAC → SECUIR)

- VERTECA's voice portal uses USC as its pipeline ontology.
- PlayNAC consumes RATE vectors without collapse, navigating its 18×7 state space via EP+GERP context (§12.4).
- SECUIR emits final cryptographic outputs using RATE vectors as authorization basis.

14. Validation Requirements (MIEVM)

All non-trivial modifications shall be validated under MIEVM across Claude, Grok, DeepSeek, and ChatGPT.

14.1 Required Validation Passes

- Isomorphism check: interlock must reduce to Triune Math #2 under all extensions.
- Conjunctive-gate check: no path may permit compensation among RATE dimensions.
- Failure-mode check: F_1–F_4 must be independently triggerable.
- Polysemantic-resolution check: primitives must return ⊥ under F_3, not the earned magnitude.
- Three-Principle check: R_7 must be queryable for every RATE vector.
- Operational-binding check (v1.2): UBIMIA, NBERS, and PlayNAC consumption paths must be exercised; 7↔7 PlayNAC-RATE correspondence must be preserved by all implementation paths.
- Uniqueness-rule check (v1.2): polysemantic primitives must have pairwise-distinct leading tokens across all readings.

15. Security Considerations

Security derives from the joint infeasibility of forging three independent conditions simultaneously.

15.1 Security Claims

Claim 15.1 (Joint Attestation Requirement). An adversary seeking to produce a RATE vector accepted by a conformant SECUIR implementation must simultaneously achieve (a) a valid CBGMODD attestation at the required tier and pillar coverage; (b) a FAVORS quorum with live Aura–ARI coupling; (c) a BERA reading sufficient to resolve the polysemantic primitives to the intent claimed. Compromise of any one condition in isolation is insufficient; compromise of any two without the third produces F_1, F_2, or F_3 respectively.

Claim 15.2 (No Intent Substitution). Declared intent alone cannot authenticate a transaction. An actor whose BERA signature does not resolve the polysemantic primitives to the declared intent will receive a RATE vector whose intent-relevant dimensions are suppressed or returned as $\perp$.

Claim 15.3 (Uncompensated Failure). RATE dimensions cannot compensate across one another. Failure on any single dimension cannot be offset by strength on others.

Claim 15.4 (Operational Binding). No abstract use of RATE is possible in conformant implementations. Every RATE vector shall be consumed by at least one of the three bound operational instruments: UBIMIA (via GCF resolution), NBERS (via BERA readings and low-R_2/R_5/R_6 triggers), or PlayNAC (via EP+GERP context navigation of the 18×7 state space). Implementations that compute RATE without routing it to an operational consumer are non-conformant. This claim prevents RATE from becoming a disembodied score.

15.2 Known Threat Classes

- **Biometric replay:** Mitigated by FAVORS Aura coupling to live BERA ARI.
- **Governance impersonation:** Mitigated by multi-pillar quorum within CBGMODD.
- **Semantic collision (F_4):** Escalated to MIEVM adjudication.
- **Intent laundering:** Structurally infeasible per Claim 15.2.
- **Abstract-scoring misuse:** Prevented by Claim 15.4; RATE cannot be used decoupled from UBIMIA, NBERS, or PlayNAC consumption.

15.3 What ERES Crypto Does Not Protect

- Wire confidentiality — use TLS-class transport.
- Coercion under duress — addressed by NPR and SECUIR at the governance layer; note that coercion patterns are partially detectable via NBERS review triggered on sustained R_6 degradation.
- Long-horizon substrate compromise — addressed by the ERES Equity Covenant.

16. Change Log and Provenance

Version	Date	Change
1.0	2026-04-21	Initial formal specification. Interlock locked. Isomorphism to Triune Math #2 established. Seven RATE dimensions enumerated. Four-mode failure taxonomy.
1.1	2026-04-21	Draft RG submission version. Added Abstract, Keywords, Related Work and Positioning, Figure 1, Worked Example with two counter-examples, three numbered Security Claims.
1.2	2026-04-21	RG submission version. Added §12 Operational Tie-Ins (UBIMIA↔GCF, NBERS↔BERA, PlayNAC↔EP+GERP). Added Figure 2 and the 7↔7 PlayNAC–RATE correspondence table. Added Claim 15.4 (Operational Binding). Added Polysemantic Uniqueness Rule (§5.5). Renamed EDF E_3 from “Earth Data Framework” to “Entity Data Framework” to eliminate duplicate leading tokens and reinforce the GAIA ENTITY architectural echo. Added normative references to NPR and SCALULAR. Added MIEVM validation passes for operational binding and uniqueness rule.

16.1 Provenance Notes

- The E_3 rename (Earth Data → Entity Data) was prompted by the polysemantic uniqueness rule surfacing “Earth” as a duplicate leading token across E_2 and E_3. The rename preserves meaning at DATA-INTEGRITY while explicitly invoking the GAIA ENTITY umbrella that unifies BODY and BRAINS.
- The 7↔7 PlayNAC-to-RATE level correspondence was observed during v1.2 consolidation. The seven-dimension RATE (introduced as a curated expansion in v1.0) aligns with the seven-level PlayNAC structure. This is registered as a design invariant in v1.2 and subject to ongoing MIEVM validation.
- Claim 15.4 (Operational Binding) formalizes a design requirement that was implicit in v1.0–v1.1: RATE must not become a disembodied score. With three operational consumers now specified, the claim can be stated precisely.
- SOMT is named in v1.2 but deferred to PlayNAC-specific specifications; the crypto stack does not require SOMT to be defined in order to specify the EP+GERP → PlayNAC interface.

17. Attestation

This specification is attested as locked and tested (“Ten”) by the author, Joseph Allen Sprute (ERES Maestro, JAS), on behalf of the ERES Institute for New Age Cybernetics. v1.2 is submitted to ResearchGate in the slot following RG #424.

Signature block:

Joseph Allen Sprute (JAS)

Founder and Director, ERES Institute for New Age Cybernetics

ORCID: 0000-0001-9946-3221

Date: April 21, 2026

Location: Bella Vista, Arkansas, USA

— End of Specification —